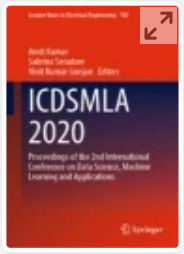
Download book PDF

Download book EPUB

Download book PDF

Download book EPUB



ICDSMLA 2020 pp 45–55

Detecting Diabetic Retinopathy Using Deep Learning Technique with Resnet-50

Viraj Jiwane, Anubhav DattaGupta, Arunkumar Chauhan & Vidya Patil

Conference paper | First Online: 09 November 2021

818 Accesses

Part of the Lecture Notes in Electrical Engineering book series (LNEE,volume 783)

Abstract

Diabetes mellitus, also known as diabetes, is a metabolic disease which may cause high blood sugar, blurred vision and several other complications. Patients already suffering from diabetes have a greater probability to develop Diabetic Retinopathy. Most prominently in the blood vessels of the tissue residing in the rear portion of the eye gets damaged causing vision problems. This disease is especially prevalent in India. Test for Diabetic Retinopathy involves a

Download book PDF Download book EPUB

which may not be easily accessible to the people especially in rural areas. Predicting the possibility of Diabetic Retinopathy utilizing fundus image is considered an effectual analysis subject by many. In this paper a method is explored to diagnose Diabetic Retinopathy using fundus images and deep learning to accurately tell the patient whether he has a Diabetic Retinopathy. Features used from the eye fundus images are soft exudate and hard exudate along with optic disc. A website is also made to deploy the model and promote mass diagnosis at a faster pace.

Keywords

Fundus images **Deep learning**

Image processing **ResNet**

Access provided by University of Liverpool
Download conference paper PDF

1 Introduction

Diabetes often leads to Diabetic Retinopathy (DR) which causes aberrations in the retina, with severe cases resulting in blindness. Most notable symptoms of this abnormality don't arise till the disease reaches a severe state. With small changes to the retinal capillaries [3]. Eventually in due time both hard exudates & retinal edema are followed

Download book PDF Download book EPUB

form hard exudates this causes clogs with the blood vessels leading to micro infarcts in the retina called exudates. When a noteworthy amount of hemorrhages, exudates or microvascular malformations are discovered, Diabetic Retinopathy is presumed to be in nonproliferative state [4]. Year 2000 saw India's population figure affected by diabetes mellitus (DM) rise to 31.7 million as stated by WHO. If the current trend is followed then by 2030 affected population is set to further rise and become 79.4 million which is by far greatest in the world [5]. While non increasing Diabetic Retinopathy in itself cannot directly affect vision it triggers macular oedema or macular ischaemia which result in rapid vision loss at any phase [6]. Early detection encourages right treatment being received which can prevent sight-threatening conditions [7]. The objective of this paper is to explore a new method of detecting Diabetic Retinopathy using deep learning specifically Resnet50 [8] making use of the features such as soft exudate, hard exudate along with optic disc.

2 Literature Survey

There has been extensive work done in the field of Computer aided diagnosis of fundus images. Many classical and adaptive methods have been applied in the recent past to detect DR. Some of the recent works with classical methods have shown an

Download book PDF Download book EPUB

Chakrabarty and Chatterjee presented a model using CNN and Relu that gave 100% accuracy [9] Shah et al. showed that Artificial Intelligence (AI) would not require help of an eye doctor for detecting presence of Diabetic Retinopathy. This would promote Remote screening and increase the amount of patients that can be scanned in a particular time period revolutionizing telescreening, especially where people have poor access to specialized health care [10]. Dharwadkar et al. in a study to identify the flowers consisting of high semantic features found that Deep Learning based neural networks gives a better accuracy in comparison with a Simple neural network [11]. Survey of Identification of Diabetic Retinopathy Detection within Fundus Images is as shown in Table 1.

Table 1 Survey of identification of diabetic retinopathy detection within fundus images. Following are the common conclusions made from the survey of recent work as shown above

Alongside detecting hard exudates, it is also essential to detect soft exudates by employing some texture analysis methods. Doing so will aid in increasing the reliability of the solution. The simple neural network is ineffective. A neural network

Download book PDF Download book EPUB

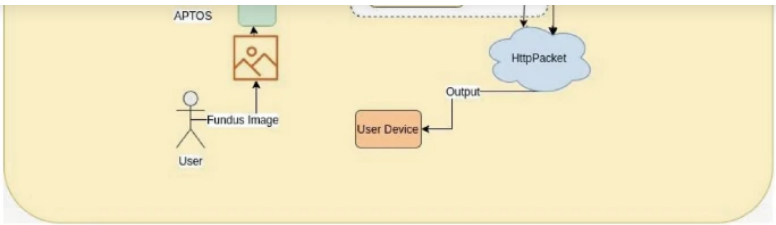
prohibits the model to generalize and learn hidden features and patterns in the dataset and so this leads to higher losses and lower probabilities of getting accurate answers. Deep neural networks face challenges when the classes in the dataset are imbalanced. The gap between the population of a certain class from other classes makes the model biased towards the said class. This challenge is faced by every application of DNN in every domain. Few papers showed high accuracy using Deep Learning Techniques using relatively small datasets.

3 Proposed System

Symptoms (like hard exudate, optic disc etc.) need to be detected as well as classified into 5 ranges of severity from the fundus image. Two different modules are employed. First is a Feature Extraction module for detecting symptoms in the image and second one is an Fundus Images Classifying module to classify the image into various severity ranges. A deep neural network is used for this purpose as shown in Fig. 1.

Fig. 1

Download book PDF Download book EPUB



High level system architecture of the proposed solution

3.1 Feature Extraction in Fundus Image

Classification: Resnet-50 is used as a base model.

Detection: For symptom detection Faster R-CNN architecture is used. Faster R-CNN is an object detection neural network architecture. This architecture is better in performance, speed and number of parameters from its predecessors because the architecture shares convolutional layers with RPN trying to learn convolutional features and thus learning the localization of the bounding boxes [20].

Algorithm

Before processing starts, images are resized down keeping the aspect ratio approximately constant. Within object detection models before actual training of the model starts, Faster R-CNN takes one of the inputs as ground truth boxes. So we need to generate these boxes using a GUI image annotation tool called Labeling. This tool saves

Download book PDF Download book EPUB

1. A classifier: Resnet 50 is used as a classifier and weights are initialized with resnet-50 coco dataset. Coco is one of the largest datasets for object detection and segmentation trained over more than 3, 30,000 images.
2. A Regional Proposal Network: it is a shallow network of 4 convolutional layers responsible for generating various regions of interest. ROI is generated using a technique called anchor boxes. Where every anchor is a pixel in the image. Various ROI's are produced centered on these anchors. Three (3) different sizes and 4 different shapes are chosen for our model. Thus 12 (3 × 4) ROIs are generated for every anchor which leads to generation of a huge number of ROIs. Before feeding them to the next network most of the ROI's are removed. By manual observation of the dataset, it is found that no image in the dataset had symptoms in more than 50 locations. So we restricted the number of boxes going in the next layer to 100, which reduces computation time for the next layer significantly.
3. A Detection layer: It takes input from both feature maps generated by classifiers and ROIs generated by RPN and tries to classify images in each ROI. Resnet-50 is chosen as a classifier for this purpose. RPN and this layer are the only 2 layers that are trained. L2 Regularizer

Download book PDF Download book EPUB

reduce the learning rate by a factor of 10 after every 1000 (approx) epochs. As an output of object detection we got bounding boxes (N × vector of size 4), labels (size × Nx1) and scores (N × 1). Bounding boxes contain diagonal corner coordinates of the rectangle localizing a symptom and the label has the prediction of which symptom was detected. And score has a probability value of the confidence with which the detection is correct.

3.2 Fundus Image Classification

The images are resized to 575 × 575 as opposed to conventional 224 × 224 because increasing the resolution a little improves the accuracy [21]. The images were given as an input into the Resnet-50 classifier, which is the most popular image classifier as it very effectively, tackles the problem of vanishing gradient by using identity connections [22]. Proposed model has a global average 2D pooling layer at end of resnet-50 network followed by a dense layer and a dropout layer of decay probability 0.5 and another dense layer and another drop out layer of decay 0.5 and a output layer of size 5 at the end [23]. All the layers are trainable. For training we used initial learning rate as 10–4 and then reduced learning rate by factor of 0.5 every time the model gets in a plateau for 3 epochs. Binary cross-entropy is used as a loss function and softmax as a final layer activation function.

Download book PDF Download book EPUB

plateau for more than 5 epochs. On prediction we got a one hot encoded vector of size 5 which contains the prediction (classified range of severity). A web application is developed with a very intuitive UI for the proposed model. The user needs to upload a fundus image and the image will be sent to the server through http request. On the server side the image gets resized to 575 × 575. Both the Feature Extraction and Fundus Images Classifying models run simultaneously with the same input image. By multithreading the time taken to process has been reduced from 5 min to 113 s.

4 Results

This section shows the results obtained by implementing the system. Figure 2 shows a graph of training loss and accuracy. We observed higher accuracy (close to 0.9) and lower loss (below 0.4) but with very slight model over fitting.

Fig. 2



Training loss and accuracy

Download book PDF Download book EPUB

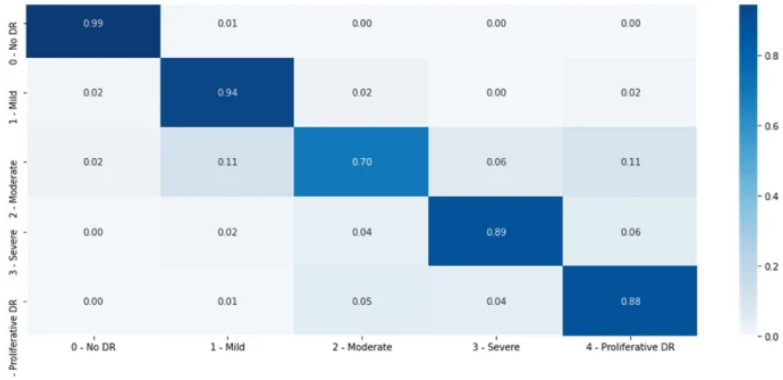
identified like soft exudate, Hard Exudate, hemorrhage, Optic Disc, Microaneurysms etc. and bounding boxes are drawn around it with its confidence. The Result of symptom identification is as shown in Table 2.

Table 2 Result of symptom detection

4.2 Disease Severity Classification

In Disease severity classification we identified the severity class to which fundus image belongs. Results of classification are as given in Fig. 3. It shows Accuracy of 0.94 over 3662 Images (Tables 3 and 4).

Fig. 3



Classification matrix

Table 3 Performance of matrices of proposed solution

Download book PDF Download book EPUB

Table 4 Balanced accuracy score of various models with different no of trainable layers

5 Conclusion and Future Scope

Paper shows implementation of a Diabetic retinopathy recognition system based on Faster R-CNN and Resnet50 for Object detection and Image classification respectively. About 94% accuracy is achieved for severity detection of 'Diabetic retinopathy'. A very user friendly and intuitive GUI is also developed for the proposed system. The GUI can be used by a patient or a doctor. Proposed model shows that diagnosis of this very dangerous disease can be brought to every household. The system takes 113 s to generate the complete result. If this can be accelerated by either better architecture, techniques, models, dataset or training time, the system can be used for real time prediction by connecting it to a fundus camera. A few things which can be worked upon are the difficulties in detecting microaneurysms and hemorrhages. Future scope also includes reducing over fitting of the model by training over a larger dataset or an ensemble of different datasets. A variation in exposure, focal length, and resolution will ultimately lead to generalization of the features of the symptoms and the image as a whole.

Download book PDF Download book EPUB

treatment, prevention, and more. Healthline, 04 Oct 2018. https://www.healthline.com/health/diabetes (accessed 16 June 2020)

2. Diabetic retinopathy—Diagnosis and treatment—Mayo Clinic, May 2018. Accessed: 16 June 2020. [Online]. Available: https://www.mayoclinic.org/diseases-conditions/diabetic-retinopathy/diagnosis-treatment/drc-20371617

3. Nentwich MM (2015) Diabetic retinopathy—ocular complications of diabetes mellitus. World J Diabetes 6(3):489. https://doi.org/10.4239/wjd.v6.i3.489

4. Saha SK, Xiao D, Bhuiyan A, Wong TY, Kanagasingam Y (2019) Color fundus image registration techniques and applications for automated analysis of diabetic retinopathy progression: a review. Biomed Signal Process Control 47:288–302. https://doi.org/10.1016/j.bspc.2018.08.034

5. Gadkari SS, Maskati QB, Nayak BK (2016) Prevalence of diabetic retinopathy in India: the all India ophthalmological society diabetic

Download book PDF Download book EPUB

6. Manjramkar M (2018) Survey of diabetic retinopathy screening methods. In: 2018 2nd international conference on trends in electronics and informatics (ICOEI). https://doi.org/10.1109/icoei.2018.8553843

7. Website. https://www.nei.nih.gov/learn-about-eye-health/eye-conditions-and-diseases/diabetic-retinopathy (accessed 16 June 2020)

8. Wu Y, Hu Z (2019) Recognition of diabetic retinopathy based on transfer learning. In: 2019 IEEE 4th international conference on cloud computing and big data analysis (ICCCBDA). https://doi.org/10.1109/icccbda.2019.8725801

9. Chakrabarty N, Chatterjee S (2019) An offbeat technique for diabetic retinopathy detection using computer vision. In: 2019 10th international conference on computing, communication and networking technologies (ICCCNT). https://doi.org/10.1109/icccnt45670.2019.8944633

10. Shah P, Mishra D, Shanmugam M, Doshi B, Jayaraj H, Ramanjulu R (2020) Validation of

Download book PDF Download book EPUB

—artificial intelligence versus clinician for screening. Indian J Ophthalmol 68(2):398. https://doi.org/10.4103/ijo.ijo_966_19

11. Dharwadkar S, Bhat G, Subba Reddy NV, Aithal PK (2017) Floriculture classification using simple neural networks and deep learning. In: 2017 2nd IEEE international conference on recent trends in electronics, information and communication technology (RTEICT). https://doi.org/10.1109/rteict.2017.8256671

12. Qummar S et al (2019) A deep learning ensemble approach for diabetic retinopathy detection. IEEE Access 7:150530–150539. https://doi.org/10.1109/access.2019.2947484

13. Chakrabarty N (2018) A deep learning method for the detection of diabetic retinopathy. In: 2018 5th IEEE Uttar Pradesh section international conference on electrical, electronics and computer engineering (UPCON). https://doi.org/10.1109/upcon.2018.8596839

14. Huda SMA, Asiful Huda SM, Ila IJ, Sarder S, Shamsujjoha M, Ali MNY (2019) An improved approach for detection of diabetic retinopathy using feature importance and machine

Download book PDF Download book EPUB

communications (ICSCC). https://doi.org/10.1109/icscc.2019.8843676

15. Jiang H, Yang K, Gao M, Zhang D, Ma H, Qian W (2019) An interpretable ensemble deep learning model for diabetic retinopathy disease classification. In: 2019 41st annual international conference of the IEEE engineering in medicine and biology society (EMBC), vol 2019, pp 2045–2048

16. Palavalasa KK, Sambaturu B (2018) Automatic diabetic retinopathy detection using digital image processing. In: 2018 international conference on communication and signal processing (ICCSP). https://doi.org/10.1109/iccsp.2018.8524234

17. Qomariah DUN, Tjandrasa H, Fatichah C (2019) Classification of diabetic retinopathy and normal retinal images using CNN and SVM. In: 2019 12th international conference on information and communication technology and system (ICTS). https://doi.org/10.1109/icts.2019.8850940

18. ur-Rehman M, ur-Rehman M, Khan SH, Abbas Z, Danish Rizvi SM (2018) Classification of diabetic retinopathy images based on

Download book PDF Download book EPUB

Intelligence (AICAI). https://doi.org/10.1109/aicai.2019.8701231

19. Arora M, Pandey M (2019) Deep neural network for diabetic retinopathy detection. In: 2019 international conference on machine learning, big data, cloud and parallel computing (COMITCon). https://doi.org/10.1109/comitcon.2019.8862217.

20. Website. https://arxiv.org/pdf/1506.01497

21. Zheng L, Zhao Y, Wang S, Wang J, Tian Q (2016) Good practice in CNN feature transfer (01 Apr 2016)

22. Wang F et al (2017) Residual attention network for image classification. In: 2017 IEEE conference on computer vision and pattern recognition (CVPR). https://doi.org/10.1109/cvpr.2017.683

23. Bhalekar M, Sureka S, Joshi S, Bedekar M (2020). Generation of image captions using VGG and ResNet CNN models cascaded with RNN approach. https://doi.org/10.1007/978-981-15-1366-4_3

Download book PDF Download book EPUB

retinopathy. Scientifica 2016:6838976.
https://doi.org/10.1155/2016/6838976

Author information

Authors and Affiliations

Department of Computer Engineering, Maharashtra Institute of Technology, Pune, India
Viraj Jiwane, Anubhav DattaGupta & Arunkumar Chauhan

School of Computer Engineering and Technology, MIT World Peace University, Pune, India
Vidya Patil

Editor information

Editors and Affiliations

BioAxis DNA Research Centre Private Ltd., Hyderabad, Telangana, India
Dr. Amit Kumar

Department of Computer Engineering, Electrical Engineering and Applied Mathematics, University of Salerno, Fisciano, Salerno, Italy
Prof. Dr. Sabrina Senatore

Department of Computer Science and Engineering, CMR Institute of Technology, Hyderabad, Telangana, India

Download book PDF Download book EPUB

Rights and permissions

Reprints and Permissions

Copyright information

© 2022 The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

About this paper

Cite this paper

Jiwane, V., DattaGupta, A., Chauhan, A., Patil, V. (2022). Detecting Diabetic Retinopathy Using Deep Learning Technique with Resnet-50. In: Kumar, A., Senatore, S., Gunjan, V.K. (eds) ICDSMLA 2020. Lecture Notes in Electrical Engineering, vol 783. Springer, Singapore. https://doi.org/10.1007/978-981-16-3690-5_5

.RIS .ENW .BIB

DOI
https://doi.org/10.1007/978-981-16-3690-5_5

Published	Publisher Name	Print ISBN
09 November 2021	Springer, Singapore	978-981-16-3689-9

Online ISBN	eBook Packages
978-981-16-3690-5	Intelligent Technologies and Robotics Intelligent Technologies and Robotics (R0)

Download book PDF

Download book EPUB

Get shareable link

Provided by the Springer Nature SharedIt content-sharing initiative

Not logged in - 138.253.175.68

University of Liverpool (2000601388) - JISC Services Ltd. (3002011430) - New JISC (3002114477) - NESLI UK (3000130459) - JISC Journal Usage Statistics Portal (JUSP) (3001139967) - JISC consortium (3991330314) - JISC Collections Brettenham House South (3000251517)

SPRINGER NATURE

© 2022 Springer Nature Switzerland AG. Part of Springer Nature.